

SCAPE Bin-Picking Evaluator - App Teaser & Testing Guide

Live Web App & PWA Link: <https://scape-evaluator-782472107063.europe-west3.run.app>

Welcome to the **SCAPE Bin-Picking Evaluator**, a premium, smart data capture tool designed to streamline technical feasibility checks and hardware selection (vision systems & grippers) for bin-picking projects.

Powered by the advanced **Gemini 2.5 Flash** model, the app uses state-of-the-art multimodal AI to automatically translate free-text descriptions and cell photos into structured project specifications.

Strategic Vision & Purpose of this Version

To scale SCAPE's sales by **10x**, we need to evaluate **100x more projects** to identify the best, most feasible applications. Because human technical resources (SCAPE evaluators) are limited, this assessment process must be automated.

This first version serves two primary purposes:

1. **Help to Integrators & End-Users:** The app guides users through the complex data capture process. It embeds SCAPE's specialized knowledge about bin-picking conditions (Yes/No/? answers) to actively **nudge the user** toward optimal "Yes" specifications (e.g. clarifying cycle times, check for entanglement, surface shininess, and CAD availability).
2. **Help to SCAPE Evaluators:** Ensure SCAPE receives complete, high-quality project data from day one, and generates an automated **AI Technical Evaluation Draft** that evaluators can use as a starting point to write their final review.

Image & Visual Consistency

- **Current Version:** Uses uploaded cell and part images to run consistency checks, verifying that the text description matches the visual evidence.
- **Future Upgrade Roadmap:** Future versions will include trained models on "good" vs. "bad" bin-picking parts/geometries to give even richer visual feedback to both the integrator and SCAPE evaluator.

The Ultimate Outcome: Better, cleaner specifications and lightning-fast response times, enabling SCAPE to capture more customers without bottlenecking human evaluators.

Key Highlights

1. Multimodal AI Auto-fill Assistant (Gemini 2.5 Flash)

- **Talk to your Project:** Describe your workspace, robot installation, or parts in plain text or voice.
- **Auto-Extraction:** The AI extracts complex parameters in real-time, auto-fills questionnaire fields, and suggests updates.

- **Smart Filtering & UX:** Displays only the modified fields in a clean accept-card with automatic text-wrapping (no clipped inputs).

2. Auto-Feasibility & Warning Badges

- When you request **AI Advice**, the system scans your draft data and highlights empty or concerning fields with colored warning markers (● or ⚠) directly next to form labels.
- Tooltips explain what is missing or suboptimal (e.g. cycle time ambitions or light interference risks).

3. Business Case & Additional Opportunities

- **ROI Mockup:** Enter shift counts and labor savings to instantly calculate project payback periods.
- **Cell Automation Scans:** A text logger and photo upload field to document other manual processes in the environment that are candidates for automation.

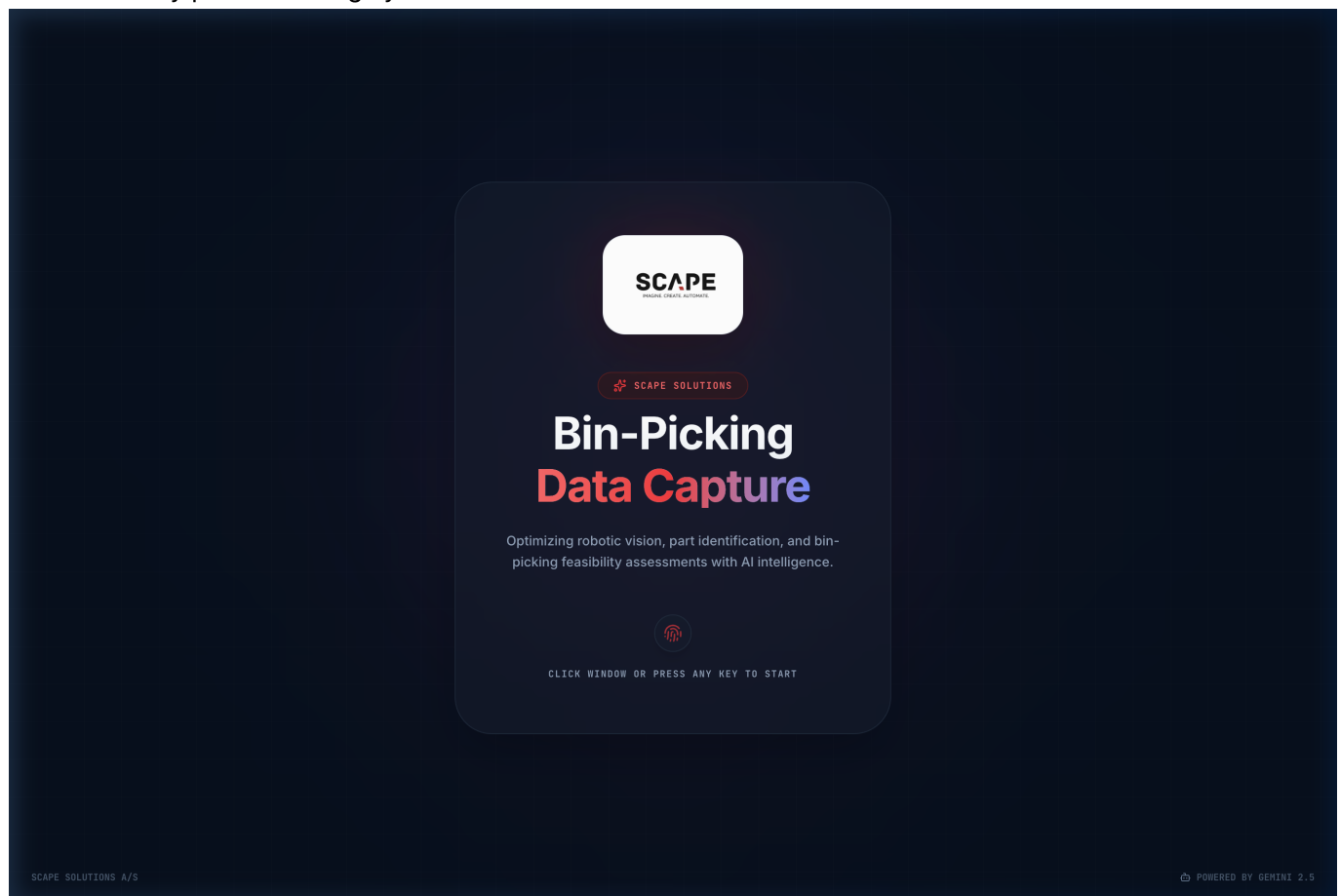
4. Seamless PWA & Mobile layout

- Fully optimized as a **Progressive Web App (PWA)** on mobile devices.
- Uses the **Visual Viewport API** to dynamically resize layouts above virtual keyboards, providing a clean chat interface with auto-scrolling questions.

Screenshots

1. App Launch & Splash Screen

Branded entry point showing system readiness.



2. Customer Project Workspace & Questionnaire (Cell Info Step)

Modular step-by-step questionnaire form with active fields, help notes, and file upload fields.

SCAPE

PROJECT: Project XXX #QPKIEI > Project & Cell Info

USER rune rune.k.larsen@gmail.com

Dashboard

STRUCTURE

Project & Cell Info 8/9

Part #1: XX

Part Dimensions Done

Characteristics Done

Visual evidence 0/1

Add Part

EXTENDED ANALYSIS

Business Case

Additional Opportunities

Review / Submit

Project & Cell Info

[1.01] Project Name

Project XXX

[1.02] Total of different parts in project

1

[1.03] Bin type

Plastic Box

[1.04_w] Bin Width (mm)

1000

[1.04_l] Bin Length (mm)

1000

[1.04_h] Bin Height (mm)

1000

[1.05] Preferred Robot Brand

KUKA

[1.06] Additional Project Notes / Info

Available cell space: Approximately 3×3 meters before the relevant machine.

[generalImages] Upload Cell & Environmental Photos OPTIONAL

Please upload photos of the robot installation area, ceiling (to check for sunlight interference), overall surroundings, or any other relevant environmental conditions.

Upload Environmental Photos

Uploaded Photos (0)

3. Questionnaire Active Warning Flags & Nudges (Part Dimensions Step)

Colored warnings (🔴 **CRITICAL**, ⚠️ **NOTE**) next to fields that do not fit "Yes" specs or are missing data (e.g. CAD availability or cycle time ambitions).

SCAPE

PROJECT: Project XXX #QPKIEI > Part #1: XX

USER rune
rune.k.larsen@gmail.com

Dashboard

STRUCTURE

Project & Cell Info 8/9

Part #1: XX

Part Dimensions Done

Characteristics Done

Visual evidence 0/1

Add Part

EXTENDED ANALYSIS

Business Case

Additional Opportunities

Review / Submit

Part Dimensions

[2.01] Part Name / Number

XX

[2.02] Part Dimensions (mm)

120mm dia x 20mm H

[2.03] Part Weight (kg)

0,03

[2.03_material] Part Material

Cardboard

[2.04] Desired Average Cycle Time (sec) NOTE

19,2

[2.05] Average Cycle Time Based On

Boxes must be emptied after each shift, with 3 boxes per shift and approximately 500 parts |

[2.06] CAD file available for the part? CRITICAL

Preferred format is STL (~0.01 mm accuracy). Other formats (STP/STEP, IGS/IGES, DWG/DXF) can be converted.

Yes

No

BACK

Continue

4. Interactive Multimodal AI Assistant Sidebar

Real-time chat where users can describe their cell. The AI parses the parameters, extracts structured facts, and applies them to the form.

SCAPE

PROJECT: Project XXX #QPKIEI > Part #1: XX

USER rune rune.k.larsen@gmail.com

Dashboard

STRUCTURE

Project & Cell Info 8/9

Part #1: XX

Part Dimensions Done

Characteristics Done

Visual evidence 0/1

Add Part

EXTENDED ANALYSIS

Business Case

Additional Opportunities

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Cardboard

[2.04] Desired Average Cycle Time (sec) NOTE

19,2

[2.05] Average Cycle Time Based On

Boxes must be emptied after each shift, with 3 boxes

[2.06] CAD file available for the part? CRITICAL

Preferred format is STL (~0.01 mm accuracy). Other formats (STP, STEP, etc.) are also supported.

Yes

for what's missing.

must be picked from the bin and placed with the open side facing upwards. The dot on one side must face forward when placed.

[2.13] Determine which side is up: Yes

[2.15] Rotational orientation must be fixed: Yes, due to a dot on one side that needs to face forward. There is a small probability of entanglement for parts in the very bottom layer in the corners, but these can be left. Only which face is up matters, but rotational orientation must also be fixed. There is only one type of part, although they come in both grey and black.

[2.07] Oil/soap/lubrication on parts: No

[2.08] Parts separated by a slip sheet: No

[2.09] Risk of entanglement: Yes, a small probability in the very bottom layer in the corners, but these can be left.

[2.10] Temperature issues with parts or environment: No

[2.11] Part very shiny or reflective: No

[2.14] Special gripper requirements: No

[1.05] Preferred robot brand: KUKA

[1.05_other] Specific KUKA robot model: KUKA KR 6 R900 sixx

[generalImages] Photos of the cell or environment available: No

[images] Photos of the part available: Yes (coming)

STILL NEED TO KNOW

1

None. All critical and flag-relevant fields have been filled.

Superseded by a newer suggestion.

change to Project XXX

AI ASSISTANT

FACTS CONFIRMED SO FAR

[1.01] Project Name: Project XXX

STILL NEED TO KNOW

1

None. All critical and flag-relevant fields have been filled.

Changes applied and saved to database.

Describe your project...

5. Automated Data Capture Advice & Project Summary

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AI feedback summarizing feasibility risks, checklist points, and details before submitting the project.

SCAPE

PROJECT: Project XXX #QPKIEI > Review / Submit

USER rune rune.k.larsen@gmail.com

Dashboard

STRUCTURE

Project & Cell Info 8/9

Part #1: XX

Add Part

EXTENDED ANALYSIS

Business Case

Additional Opportunities

Review / Submit

Data Capture Advice

HIDE ADVICE

1. QUICK ACTION SUMMARY

To help Scape provide a thorough evaluation, please prioritize uploading photos of the part, as no CAD file or images are currently available. Also, provide more detail on the required placement accuracy for the part.

Here's a direct checklist of what would be most helpful:

• Upload photos of the "Cardboard cover" part: Include parts both inside the bin (a full view and a close-up) and on a flat surface from multiple angles.

• Clarify placement accuracy for "Cardboard cover": Provide a target placement accuracy in mm, and indicate if the "architecture" (fixture) is already fixed or still flexible in its design.

• Clarify cycle time basis for "Cardboard cover": Confirm if the 19.2-second cycle time is an average target over a shift/bin, and if occasional slower cycles are acceptable.

2. DETAILED ANALYSIS

Missing or incomplete fields

• Part 1: Cardboard cover

CAD file / Part photos (2.06): Since no CAD file is available and no photos have been provided, uploading images of the part is critical. Please provide clear photos of the "Cardboard cover" part. This is essential for Scape to understand the geometry and surface characteristics for a proper evaluation.

Cycle time

• Part 1: Cardboard cover (2.04, 2.05)

You've indicated a cycle time of 19.2 seconds and noted that "Boxes must be emptied after each shift, with 3 boxes per shift and approximately 500 parts per box." This helps clarify the context. To refine this further, could you confirm if 19.2 seconds is an average target

GET ADVICE ON DATA

Parts Summary

Part #01: XX

0 Images

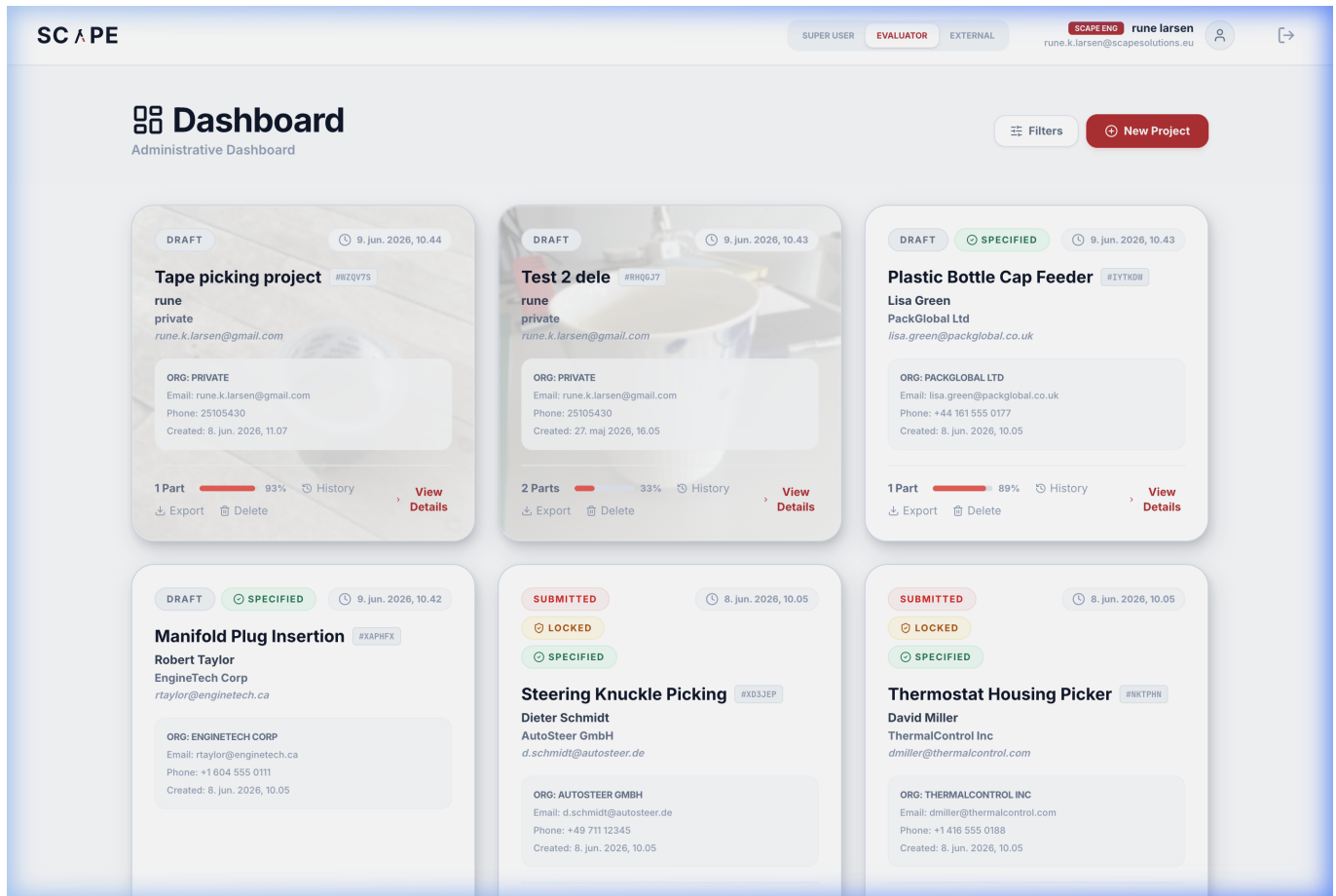
Submit to Scape Solutions

Mark as Fully Specified

6. Evaluator Administrative Dashboard

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Shows the case overview page where SCAPE evaluators can take cases, manage locks, review AI evaluator drafts, and submit verdicts.



Easiest Way to Test It

To ensure a successful demo during your meeting tomorrow, here is how to log in and what to expect based on roles:

Option A: Pre-Registered Demo Account (Highly Recommended)

Use our pre-configured demo credentials which work instantly on **all mobile web, desktop, and standalone PWA apps**:

- **Email:** demo@scapesolutions.eu
- **Password:** [ScapeEvaluator2026](#)
- **What to Expect (Admin View):** This account has been whitelisted as a **SCAPE Evaluator/Admin**. Logging in with this account lets you toggle between **User Mode** (Customer View) and **Evaluator Mode** (Admin View) in the top header. You will be able to see and manage all customer projects, write verdicts, and test the Prompts Editor.

Option B: Direct Google Sign-In for SCAPE Evaluators

If you are logging in as a SCAPE Employee, the following team accounts are already pre-qualified and whitelisted in Firestore:

- **René Dencker Eriksen:** rene.dencker.eriksen@scapesolutions.eu

- **John Erland Østergaard:** john.erland.oestergaard@scapesolutions.eu
- **Rune Klausen Larsen:** rune.k.larsen@scapesolutions.eu
- **How to log in:** Simply click the "**Sign in with Google**" button on the sign-in screen and log in using your respective @scapesolutions.eu Google Workspace account.
- **What to Expect (Admin View):** You will be authenticated immediately without needing a password. The system will match your verified email domain/address and log you in directly with full **Evaluator / Admin** rights.

Option C: Registering a New Account (Customer Isolation View)

- **Works on:** Sign up with email/password or Google Sign-In.
- **Requirements:** The email domain must end in a whitelisted domain (e.g. @scapesolutions.eu or @scapesolutions.com) or be added manually to the whitelisted emails list in Firestore.
- **What to Expect (Customer View):** Any new account will log in as a **Customer / End User** by default. To show off multi-tenant security, this user will have a clean, blank dashboard and **cannot** see anyone else's projects or access admin tools.